Pacmann

Credit Risk Analysis & Sales Optimization Simulation

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Outline

- Introduction (Problem, Objective, and Metrics)
- Dataset
- Findings (Tableau Public Link, Viz, and Insights)
- Summary/Conclusion
- Recommendations
- References

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Introduction

- Problem
  there is a critical imbalance in setting income thresholds, overly stringent policies result in significant Opportunity Loss by rejecting creditworthy applicants, while excessively lenient criteria lead to higher Default Rates, threatening the company's financial stability. This is further complicated by a lack of precise risk mapping across demographic profiles (education and occupation), making credit decisions suboptimal.

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Introduction

- Objective
  To optimize credit disbursement policies by dynamically adjusting income thresholds through data-driven simulations, aiming to achieve a balanced ratio between Approval Rates and Risk Tolerance within the current portfolio review cycle.

- Metrics
  Monitoring Approval Rate to ensure sustainable business growth, alongside Default and Delinquency Rates to control credit quality and identify high-risk occupational segments.

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Dataset

- application_record.csv & credit_record.csv

  https://drive.google.com/drive/folders/10Q1TEeSKV6kO5GJb4zvfVk1vo2OyYo9K

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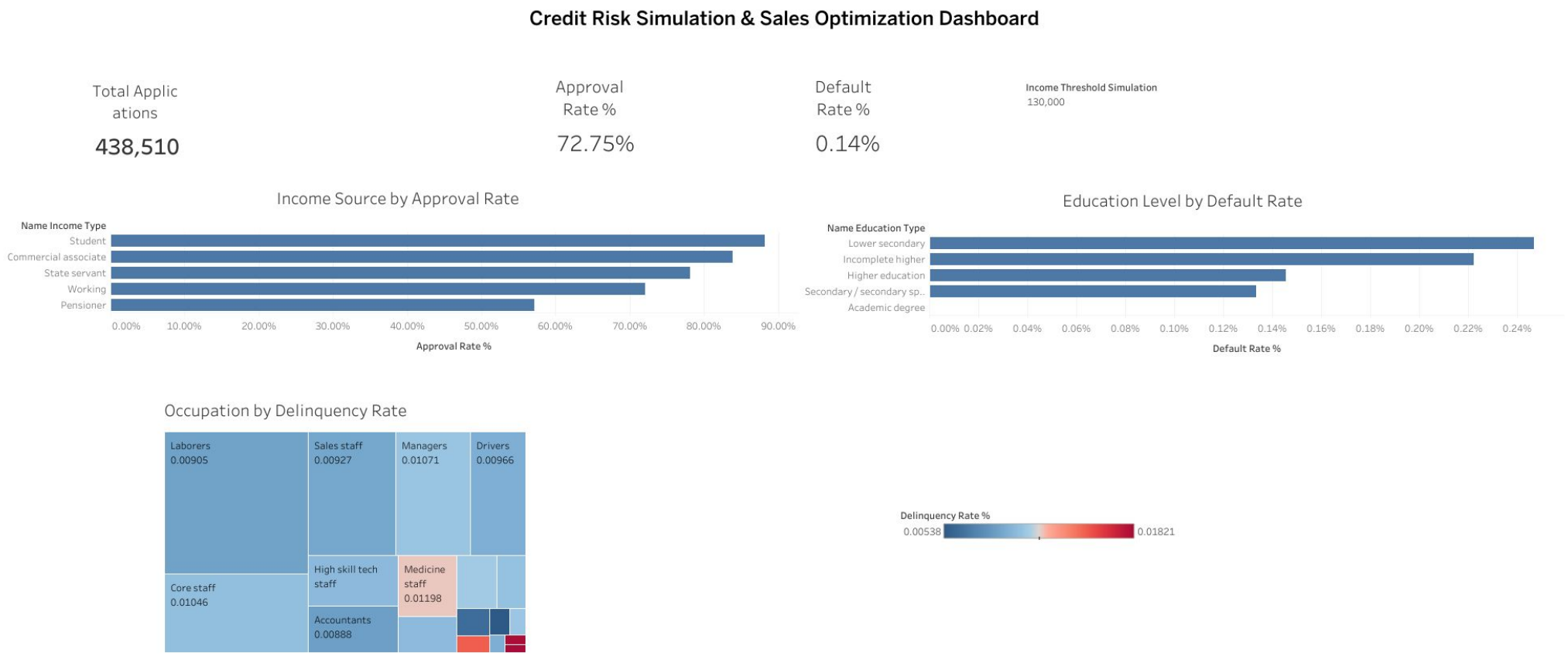
Findings



[Tableau Dashboard Link]

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Insight

- The Income Threshold acts as the primary business "faucet." Setting it too low (20,000) allows for a 100% Approval Rate (everyone is accepted), which is high-growth but high-risk. Setting it at 130,000 hits the "sweet spot", allowing a healthy 72.75% Approval Rate while keeping the Default Rate extremely low at 0.14%.

- The Data shows a direct correlation between education and repayment discipline. Lower Secondary graduates have the highest Default Rate, while those with an Academic Degree are the safest group. This proves that educational background is a powerful predictor of credit quality.

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Conclusion & Recomendations

- The best strategy is not to reject as many people as possible, but to find a moderate income threshold that allows the business to grow without sacrificing the company's financial security.

- Apply more flexible criteria for the Higher Education segment and Commercial Associates, as they consistently demonstrate the cleanest risk profiles.

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References

- Pacmann Bright Future Program Materials.
- Tableau Documentation for Simulation Dashboards.

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